

Computational Graphic Statics

Dr. Lluís Enrique Monzó
Block Research Group (Prof. Dr. Philippe Block)



BRG

<http://block.arch.ethz.ch>

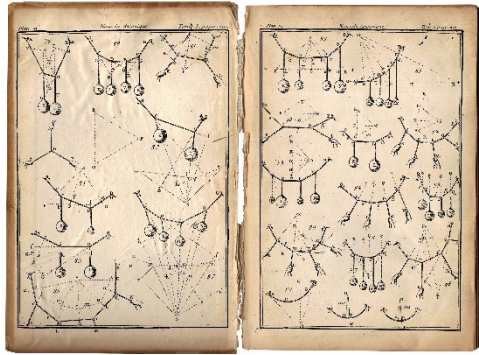


@blockresearchgroup

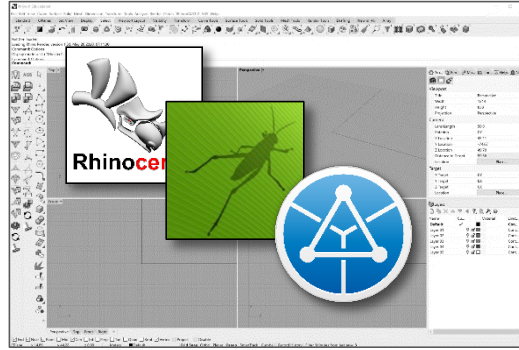


BRG





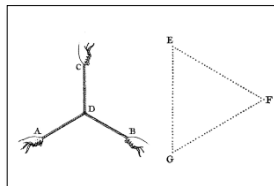
+



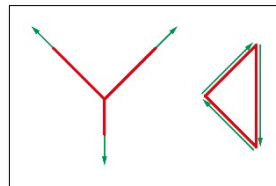
→



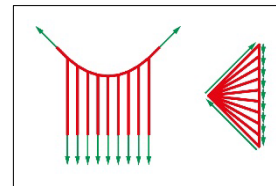
Explore new **structural design opportunities**
by combining **graphic statics** with **computational tools**



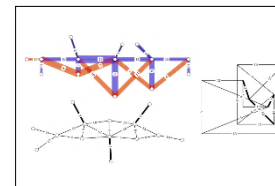
Module A
Introduction to Graphic Statics



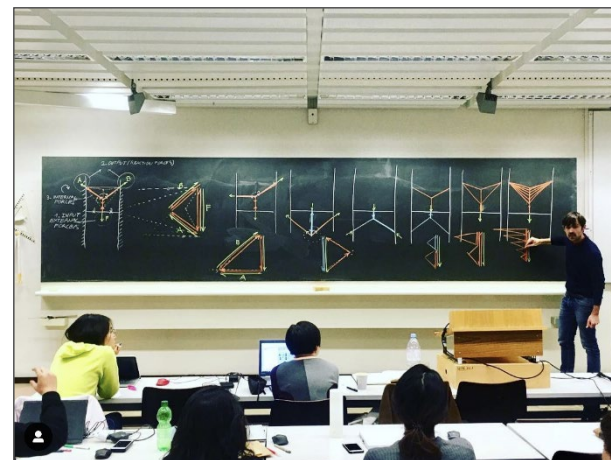
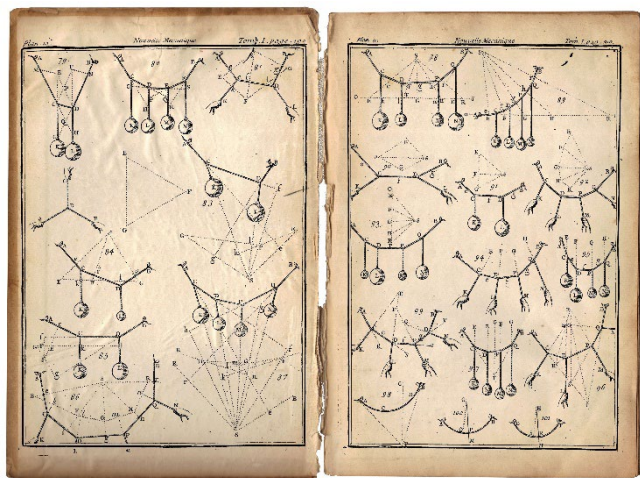
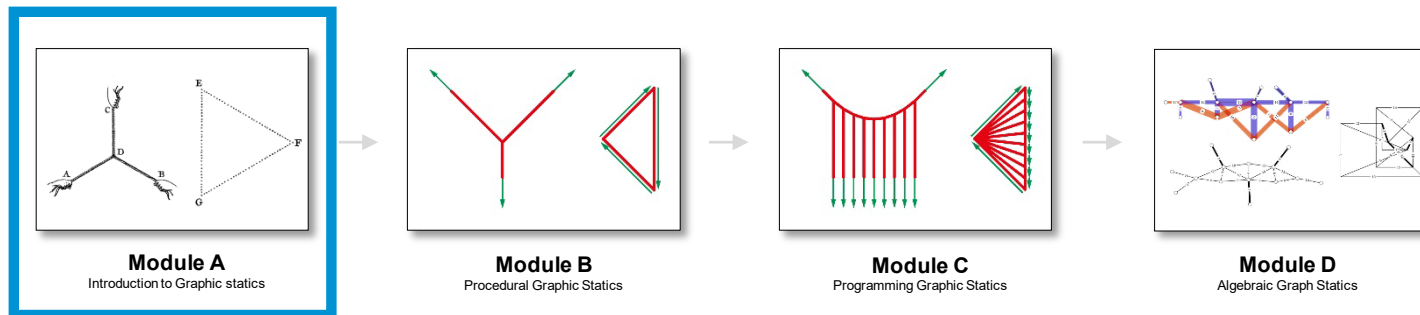
Module B
Procedural Graphic Statics

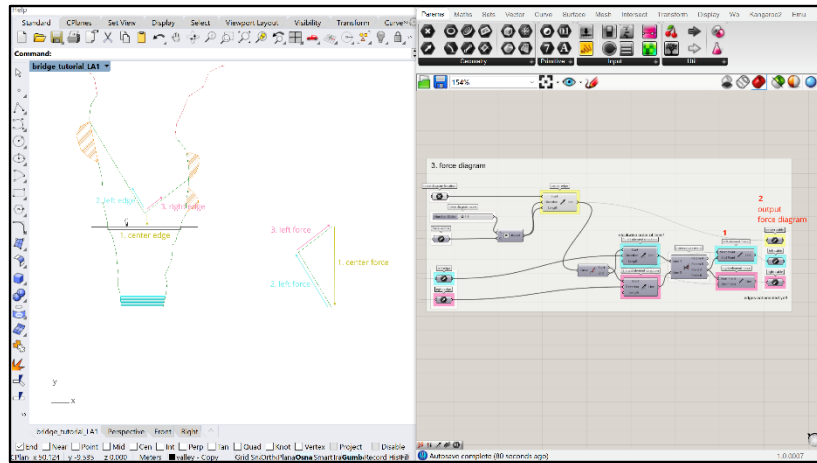
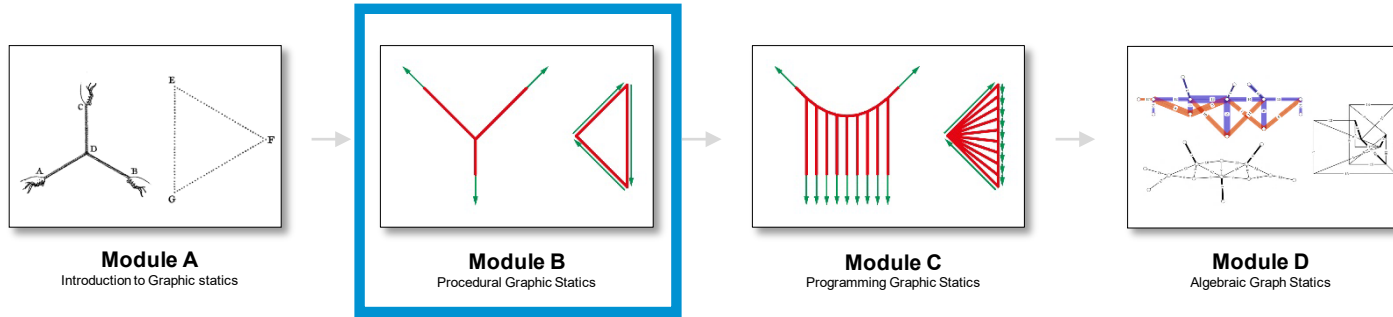


Module C
Programming Graphic Statics



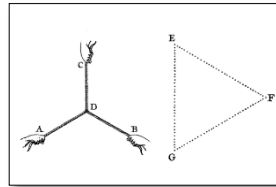
Module D
Algebraic Graph Statics



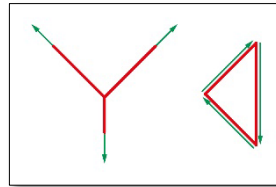


+

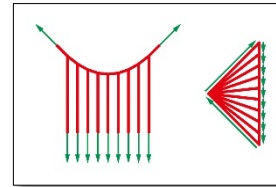




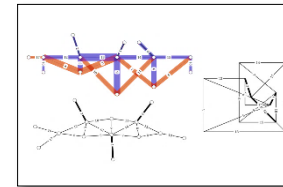
Module A
Introduction to Graphic statics



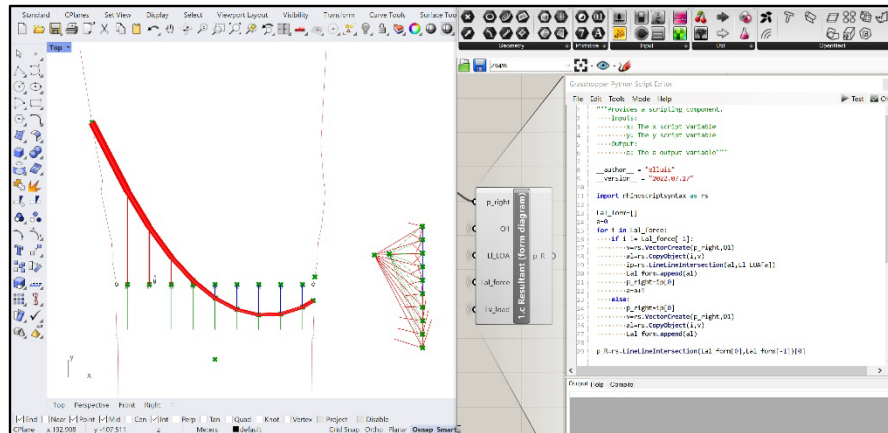
Module B
Procedural Graphic Statics



Module C
Programming Graphic Statics



Module D
Algebraic Graph Statics

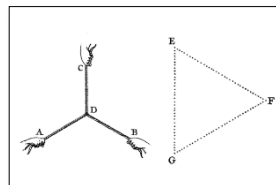


+

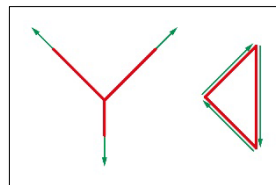


+

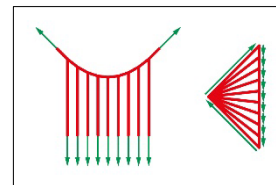




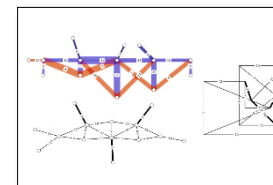
Module A
Introduction to Graphic statics



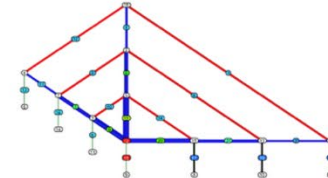
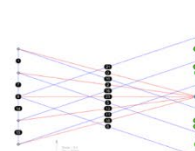
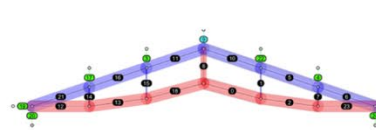
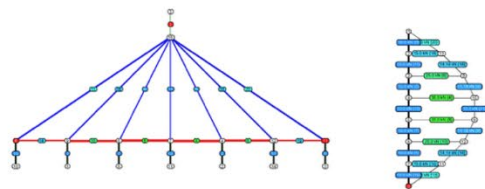
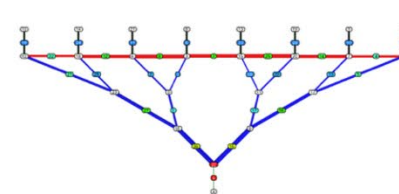
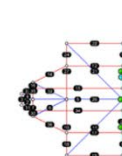
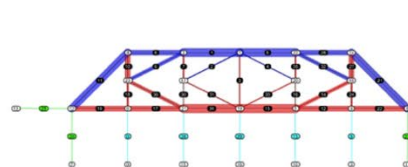
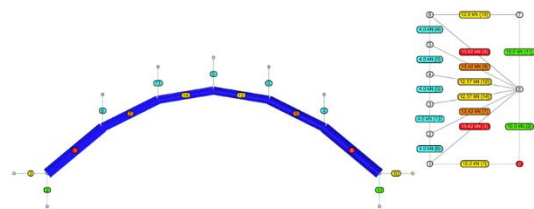
Module B
Procedural Graphic Statics

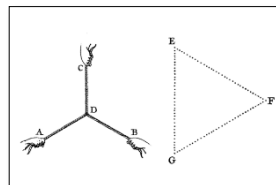


Module C
Programming Graphic Statics

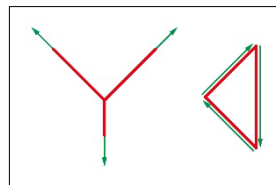


Module D
Algebraic Graph Statics

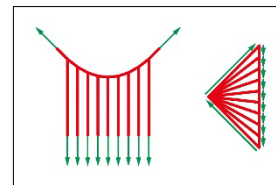




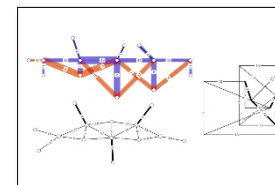
Module A
Introduction to Graphic statics



Module B
Procedural Graphic Statics



Module C
Programming Graphic Statics



Module D
Algebraic Graph Statics



compas_ag

By the end of the course, students should be able to:

- Demonstrate basic use of graphic statics for form-finding and structural analysis.
- Understand the goals and structure of an algorithm.
- Develop algorithms based on graphic statics to explore equilibrium systems.
- Explain the potential of using computational tools to study graphic statics models.
- Use Interactive Graphic Statics (IGS2) for the analysis, form-finding, and design of 2D structures.
- Apply computational graphic statics in design contexts.
- Demonstrate basic skills in Rhinoceros, Grasshopper and Python scripting.

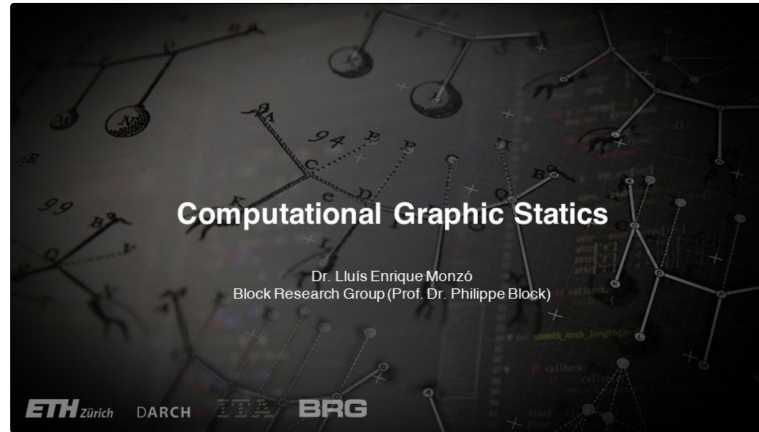
A	Week 1 (20/9)	Lecture	Introduction to graphic statics
		Tutorial	Fundamentals of graphic statics Introduction to Rhino and Grasshopper
		Intro exercise	EX A-1: Graphic statics in Rhino EX A-2: Grasshopper basics
B	Week 2 (27/9)	Submission	EX A-1 and EX A-2
		Lecture	Algorithmic design & thinking
		Tutorial 1	Single node bridge
	Week 3 (04/10)	Intro exercise	EX B-1 : Single node bridge
		Work session	
	Week 4 (11/10)	Submission	EX B-1
		Tutorial 2	Form finding of funicular structures
	Week 5 (18/10)	Intro exercise	EX B-2 : Form finding of funicular structures
		Work session	
Week 6: seminar week			
C	Week 7 (01/11)	Submission	EX B-2
		Lecture	Computational Graphic Statics
		Tutorial 1	Programming in Grasshopper Python
		Intro exercise	EX C-1: Programming in Grasshopper Python
	Week 8 (08/11)	Submission	EX C-1
		Tutorial 2	Form finding of funicular structures in Grasshopper Python
	Week 9 (15/11)	Intro exercise	EX C-2: Form finding of funicular structures in Grasshopper Python
Work session		EX C-2	
D	Week 10 (22/11)	Submission	EX C-2
		Lecture	Algebraic Graph Statics (AGS)
		Tutorial 1	Analysis of trusses using Interactive Graphic Statics (IGS2)
		Intro exercise	EX D-1: Analysis of trusses using Interactive Graphic Statics (IGS2)
	Week 11 (29/11)	Submission	EX D-1
		Tutorial 2	Form-finding using Interactive Graphic Statics (IGS2)
		Intro exercise	EX D-2: Form-finding using Interactive Graphic Statics (IGS2)
	Week 12 (06/12)	Work session	EX D-2
Submission		EX D-2	

[Ask or Search](#) **Ctrl + K**[About](#)[Syllabus](#)

COURSE MATERIAL

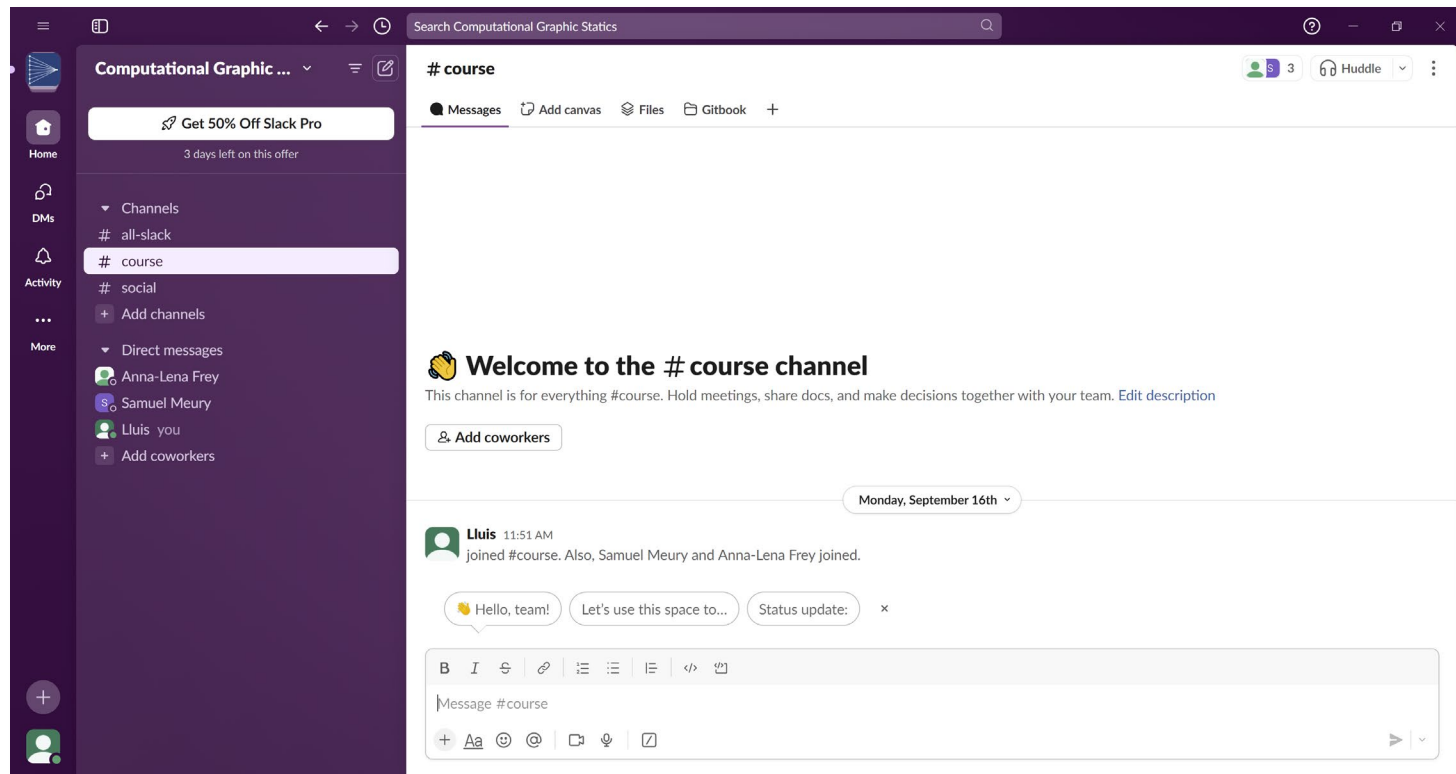
[A. Introduction](#)[Lecture](#)[Tutorial](#)[Exercise](#)

About

[Welcome to Computational Graphic Statics!](#)[General information](#)[Export as PDF](#)

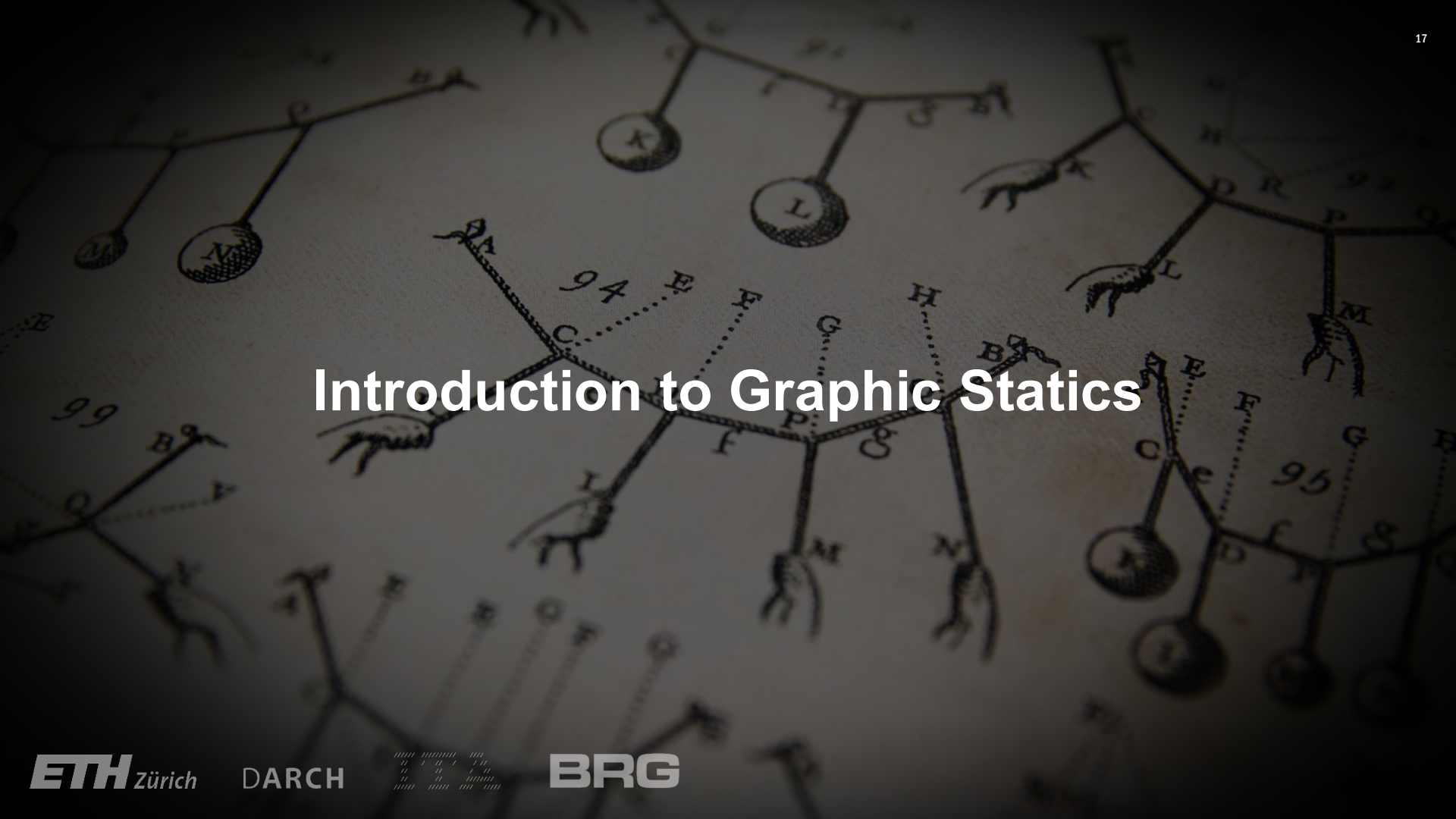
Welcome to Computational Graphic Statics!

This course explores new structural design opportunities that arise when combining graphic statics—an intuitive, equilibrium-based method for form-finding, analysis, and design of structures—with computational tools. The course begins with a review of graphic statics fundamentals and an introduction to basic parametric tools. It then focuses on various computational approaches for programming graphic statics algorithms, ranging from linear (procedural) algorithms to more complex data structures.

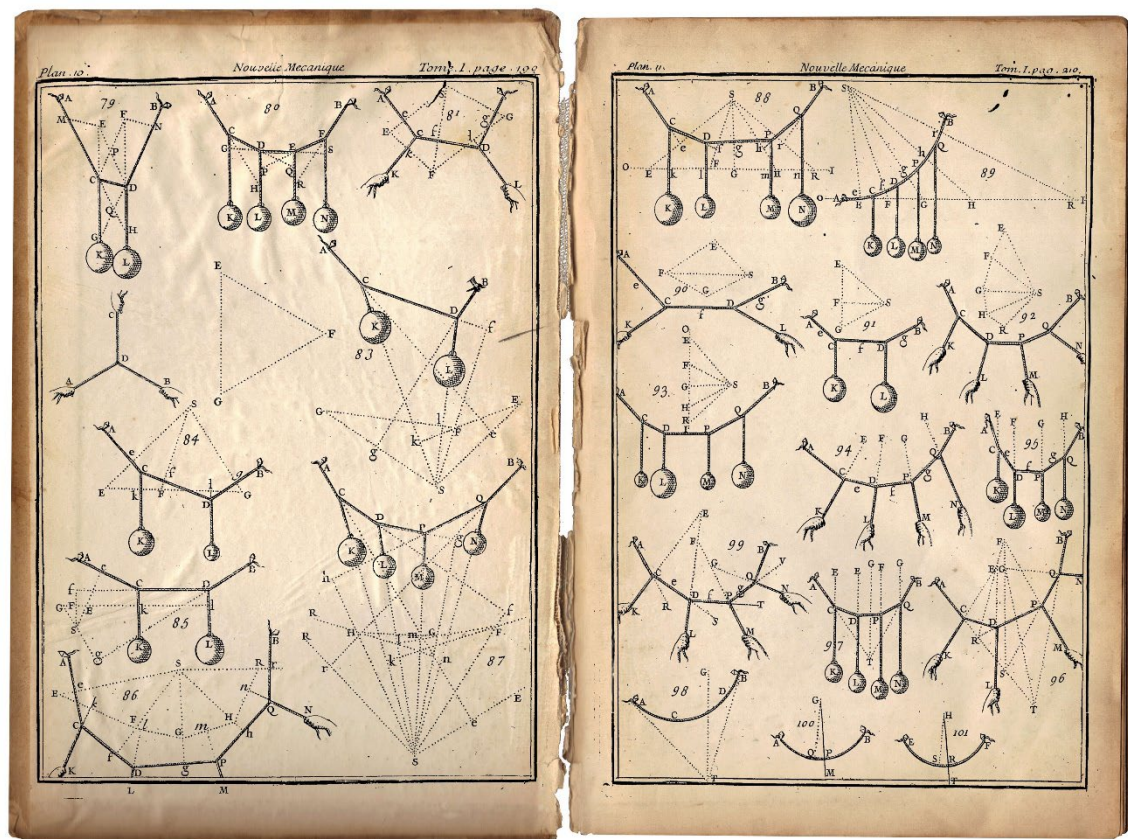


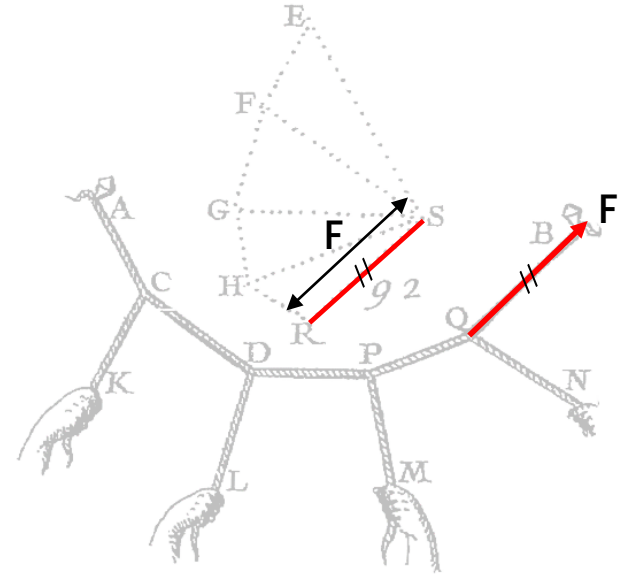
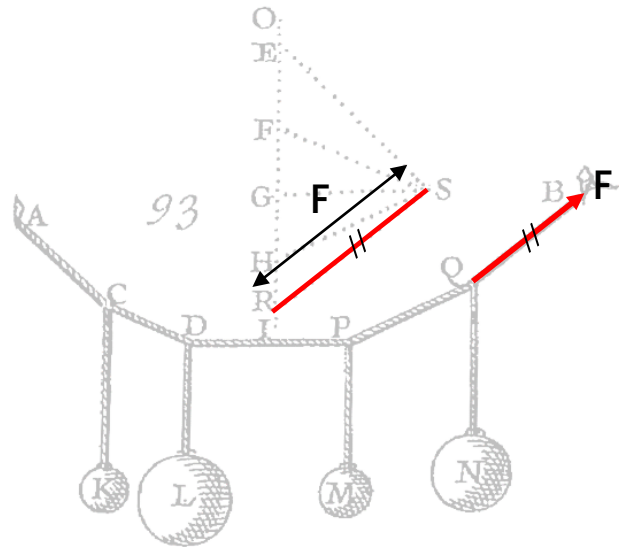
Week 1

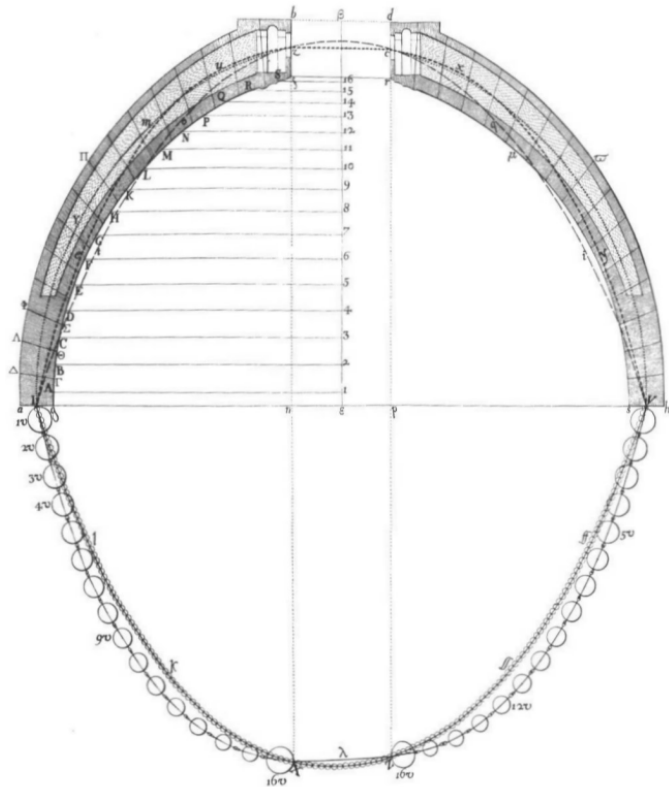
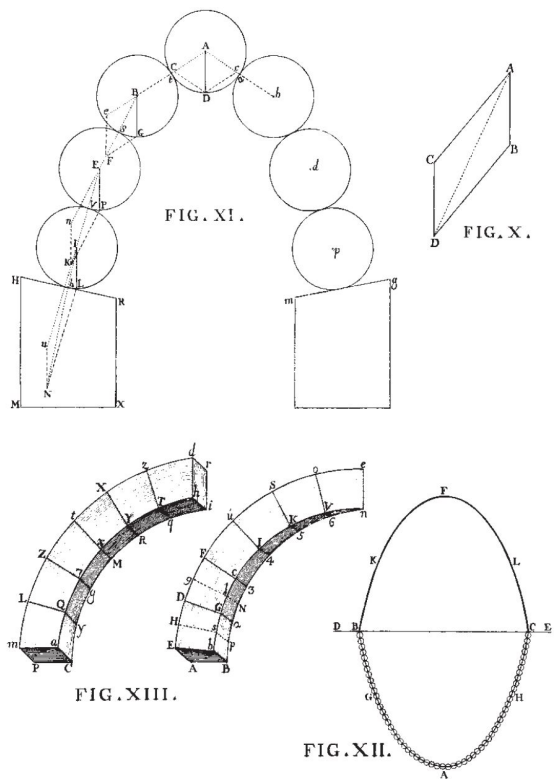
9:45 – 10:30	Welcome & introduction Lecture: Introduction to Graphic Statics
10:30 – 10:45	Break
10:45 – 11:30	Tutorial: Fundamentals of Graphic Statics Tutorial: Introduction to Rhinoceros
11:30 – 11:45	Break
11:45 – 12:30	Tutorial: Introduction to Grasshopper Introduction exercises

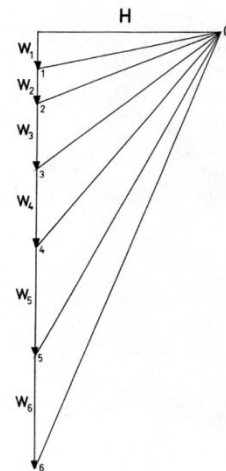
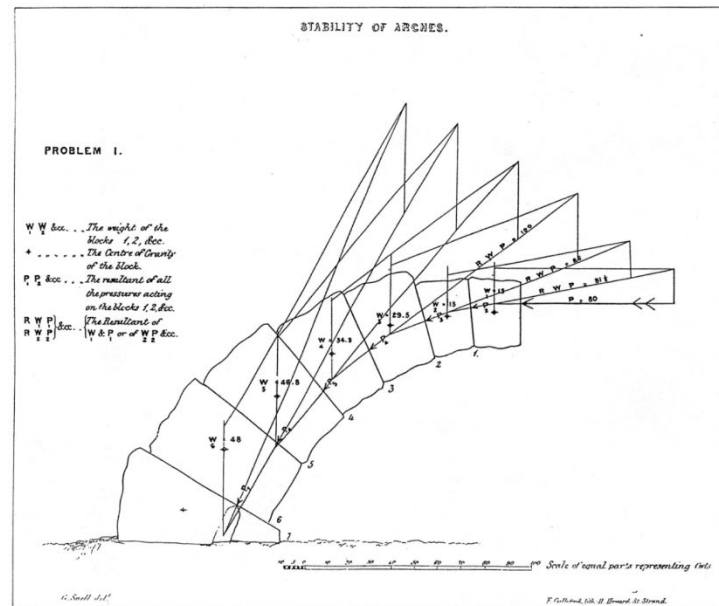
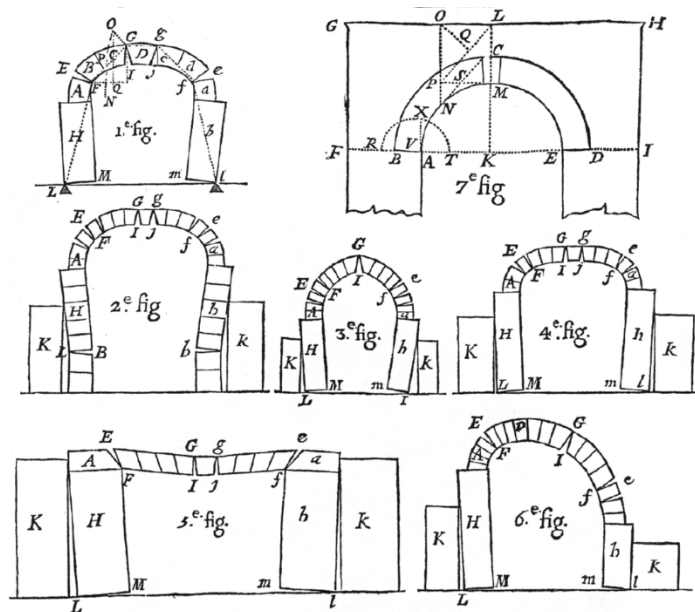


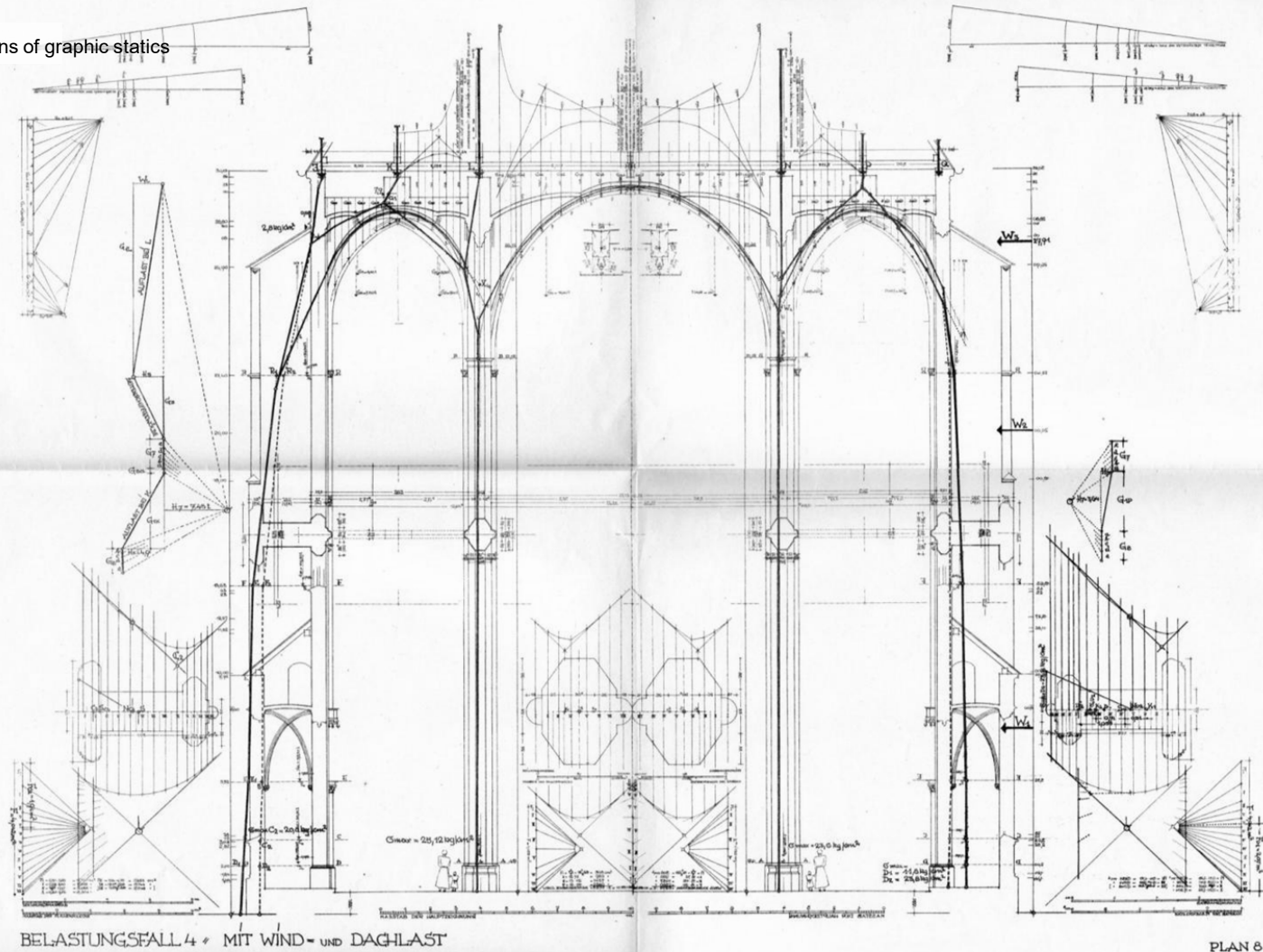
Introduction to Graphic Statics

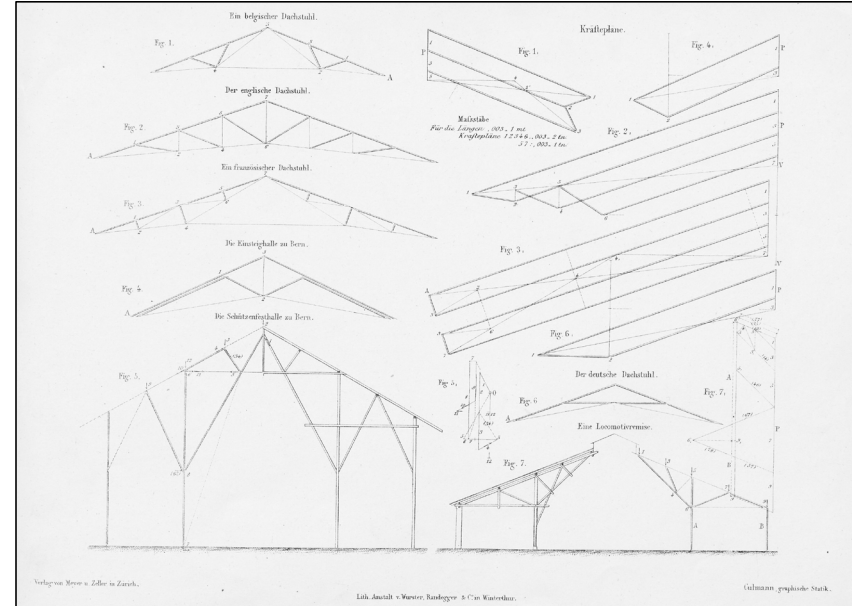
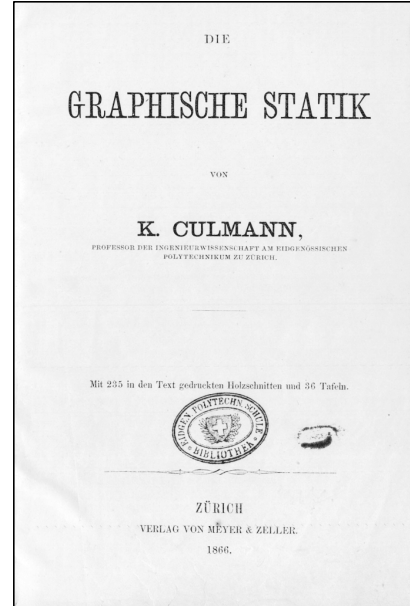












156 EXAMPLES OF FRAME- AND STRESS- DIAGRAMS. [45-

the upper extremity of the segment 1 meets the upper extremity of the segment 2; and the straight line 18 through the point (16, 1), which is both the lower extremity of the segment 16 and that of the segment 1.

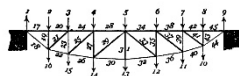


Fig. 8a.

Pass on to the joint (2, 17, 19, 30). Draw 19 through the point (17, 18), and 20 through the point (2, 3), the lower end of 2 and upper extremity of 3; and we obtain the polygon 2, 17, 19, 20, which is a rectangle.

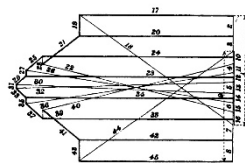


Fig. 8b.

Construct the polygon corresponding to the joint (16, 18, 19, 21, 22). For this purpose draw 21 through the point (19, 20), and 22 through the point (15, 16); we thus obtain a crossed pentagon. Continue to deal in the same manner with each of the points of application of the forces 3, 15, 4, 14, 13, 6, 12, 6, 11, 7, 10, 9, taken in succession.

Since the diagram *a*, which represents the skeleton of the structure and all the external forces, has for its axis of symmetry the vertical which passes through the centre of the figure, the diagram *b* has for its axis of symmetry the median horizontal line. For example, the triangle 9, 45, 44 is sym-

-44] EXAMPLES OF FRAME- AND STRESS- DIAGRAMS. 157

metrical to the triangle 1, 17, 18; the rectangle 8, 45, 43, 42 to the rectangle 2, 17, 19, 20; and so on.

All the upper bars are in compression, and all the lower ones are in tension.

The diagonals and contra-diagonals are all in compression; finally two of the verticals 23, 39 are in tension, and all the rest in compression.

44. Figure 9a* represents one half of a locomotive shed. The external forces are the weights 1, 2, 3, 4, 5 applied at

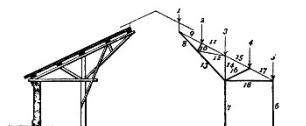


Fig. 9a.

the upper joints of the frame, and the reactions 6 and 7 of the wall and column. Again, all the external forces are parallel, and consequently the polygon of forces reduces, in diagram *b*, to one straight line. The force 6 (taken in the opposite sense to that in which it really acts) is equal to a certain part of the weight 5; by adding the difference to the other weights we get the magnitude of the force 7.

In the diagram *b* the direction of the lines 8 and 13 coincide; the first is a part of the second. Here then we have for the polygon corresponding to the joint (8, 10, 12, 18) one of those degenerate forms about which we spoke in Art. 33; the polygon is in fact a quadrilateral 8, 10, 12, 13, having three of its vertices (13, 9), (8, 10), (12, 13) in one straight line.

The polygon 5, 17, 18, 6, corresponding to the point where

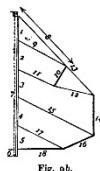


Fig. 9b.

* This example is taken from PLATE XIX. of the *Atlas of Graphic Statics* of CULMAN, 1st edition. As previously stated, the two diagrams are not rigorously reciprocal.

158 EXAMPLES OF FRAME- AND STRESS- DIAGRAMS. [45-

the roof is supported by the wall, presents an analogous degenerate form, since the vertices (6, 5), (upper point of the segment (6), (5, 17), and (18, 6) all lie in the same straight line.

The lower bars 8, 13, 18 are in compression, as well as the diagonals 10, 14, 16, the column 7 and the wall 6; while the upper pieces 9, 11, 15, 17 and the diagonal 12 are in tension.

45. Diagram *a* of Fig. 10 represents a truss at the upper joints of which are applied the oblique forces 1, 2, ..., 7, which

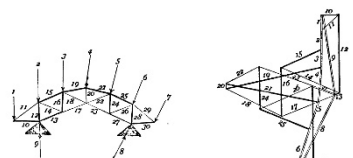


Fig. 10a.

may be considered as the resultants of the dead-loads and wind pressure; the forces 8, 9 represent the reactions at the supports.

The polygon of external forces is drawn in diagram *b* with double lines.

We construct successively the triangle 1, 10, 11, the quadrilateral 9, 10, 12, 13, the pentagon 2, 11, 12, 14, 15, the quadrilateral 13, 14, 16, 17, the crossed pentagon 3, 15, 16, 18, 19; the crossed quadrilateral 4, 19, 20, 21, the pentagon 17, 18, 20, 22, 23, and so on.

The upper bars 16, 19, 21, 26 are in compression, as well as the lower bars 10, 13, 20, and the verticals 12, 16, 24, 28; whilst all the remaining bars of the structure are in tension.

46. The diagram *a* of Fig. 11 represents a suspension bridge, loaded at each of its upper joints with weights 1, 2, ..., 8, and at each of its lower joints with weights 10, 11, 21, ..., 16; the weights are kept in equilibrium by the two

-46] EXAMPLES OF FRAME- AND STRESS- DIAGRAMS. 159

oblique reactions 9, 17 at the two extreme points of the structure*.

The polygon of external forces has its first eight sides in succession along the same vertical straight line, and its seven

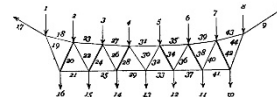


Fig. 11a.

last sides situated in another vertical straight line. The oblique sides 9 and 17 intersect, so that the polygon is a crossed one. We construct successively the polygons 1, 17, 19, 18; 16, 19, 20, 21; 2, 18, 20, 22, 23; 15, 21, 22, 24, 25; 3, 23, 24, 26, 27; and so on; most of which are crossed.

Diagram *b* shows that the upper bars are all in tension, and that the tension decreases from the ends towards the

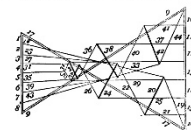


Fig. 11b.

middle of the structure; the bars of the lower boom are also all in tension, but in them the tension decreases from the middle towards the ends.

The extreme diagonals and contra-diagonals are in tension; in the portion situated to the left of the axis of symmetry, the diagonals or braces are alternately in tension and compression; similarly they are on the right but in the reverse order. Considering separately the ties and struts, we see that the internal

* This example is analogous to one of those studied by Maxwell in his memoir of 1870.

PLATE VII.

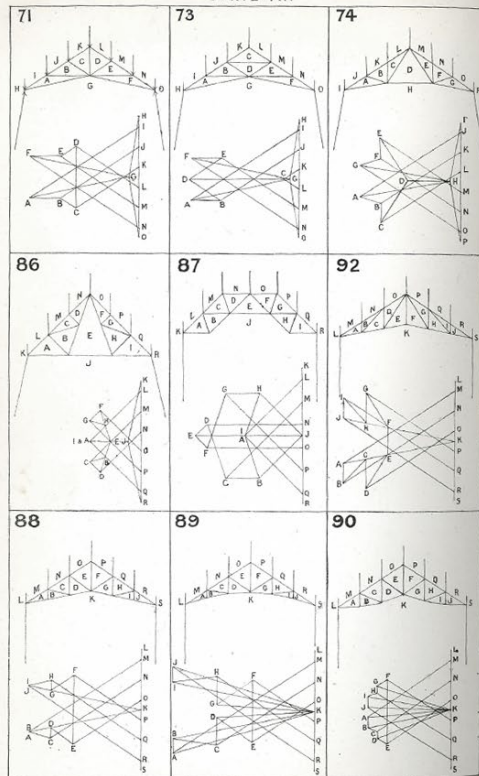
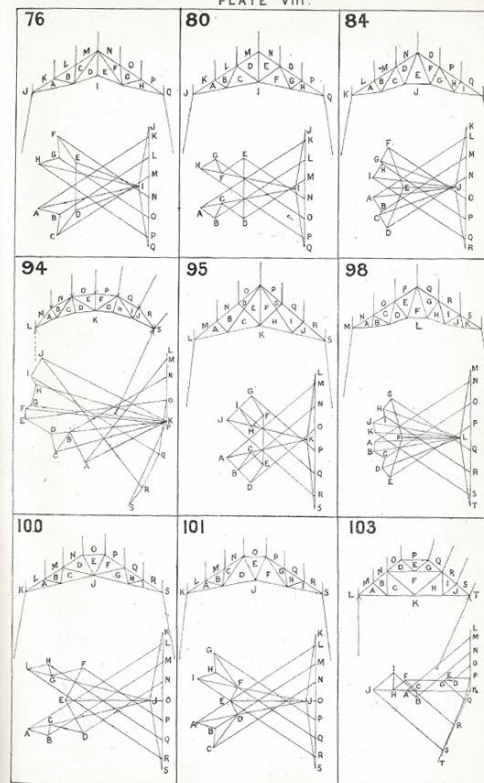
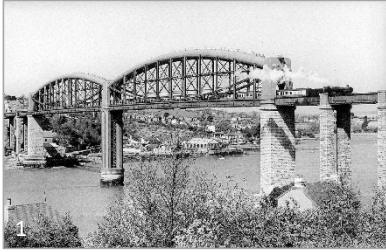


PLATE VIII.





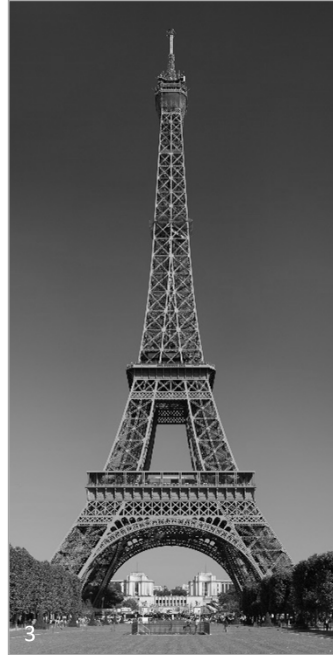
Royal Albert Bridge (1859)

Isambard Kingdom Brunel



Garabit Viaduct (1884)

Maurice Koechlin



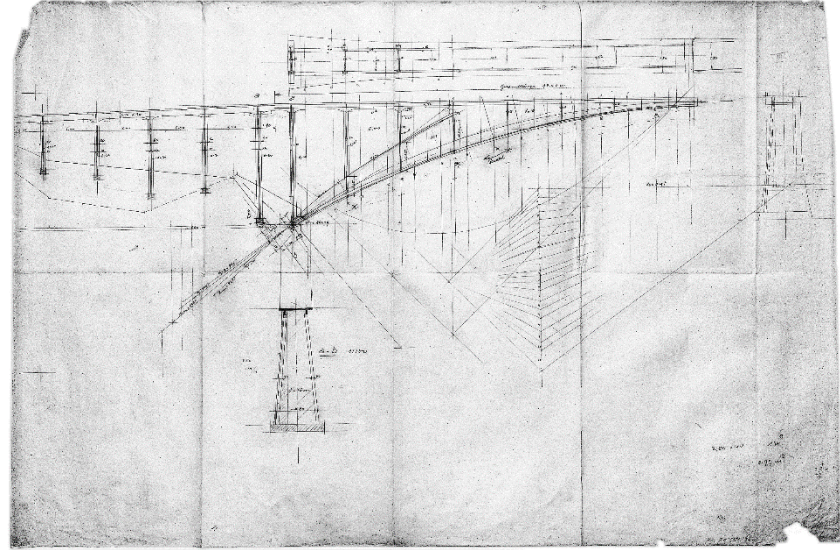
Eiffel Tower (1887)

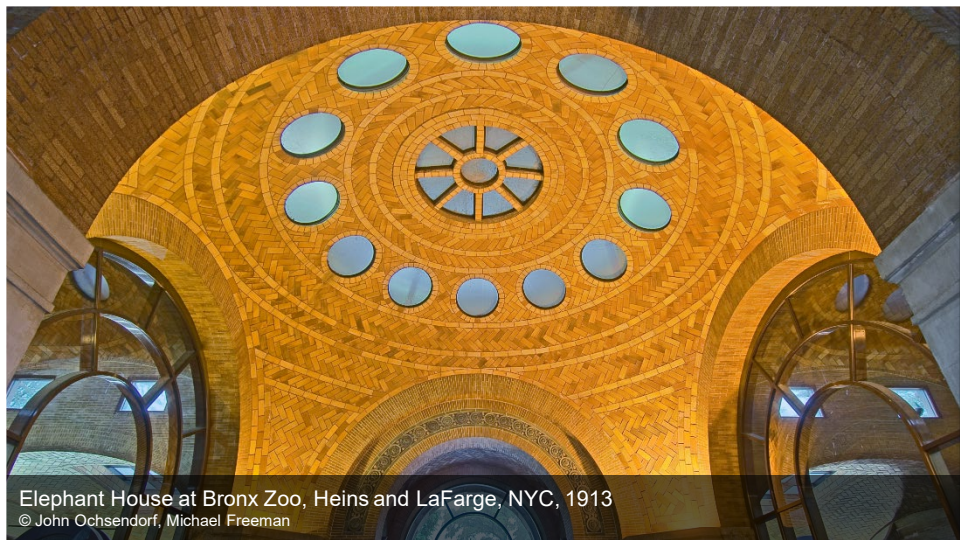
Maurice Koechlin, Émile Nouguier

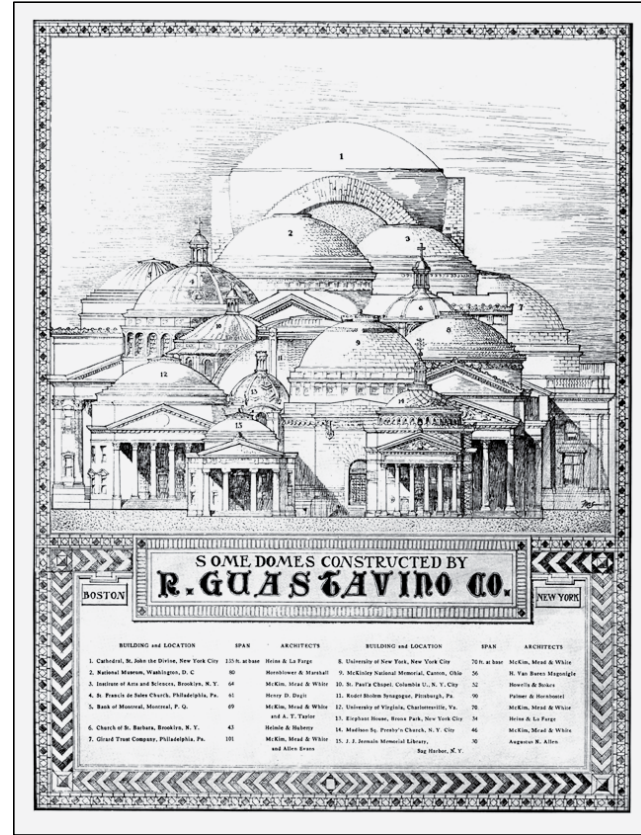
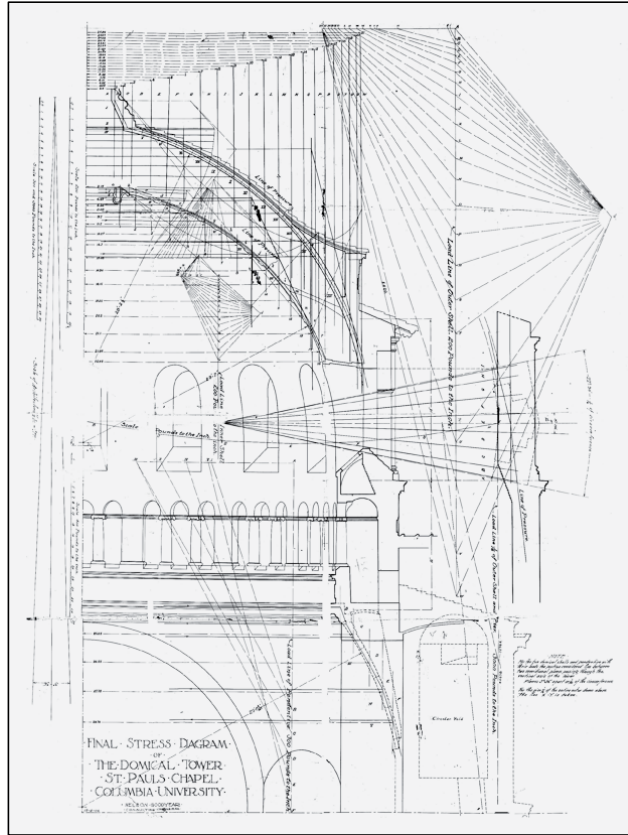


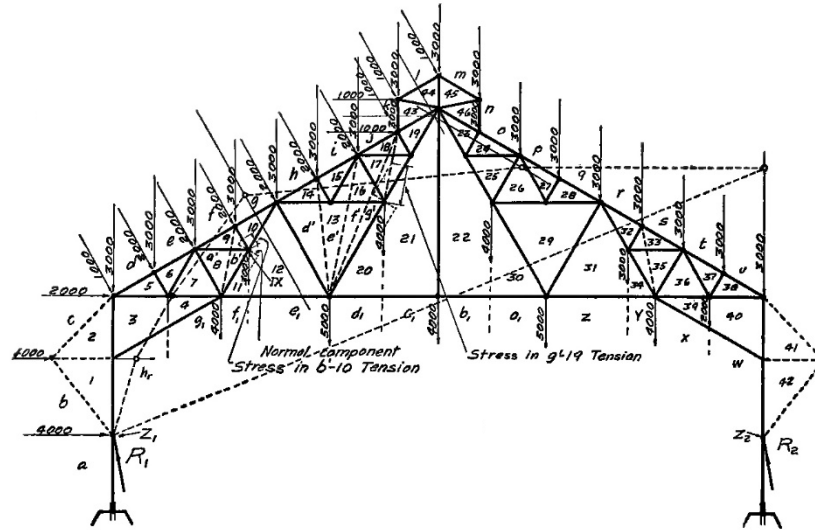
Forth Bridge (1890)

Sir John Fowler, Sir Benjamin Baker

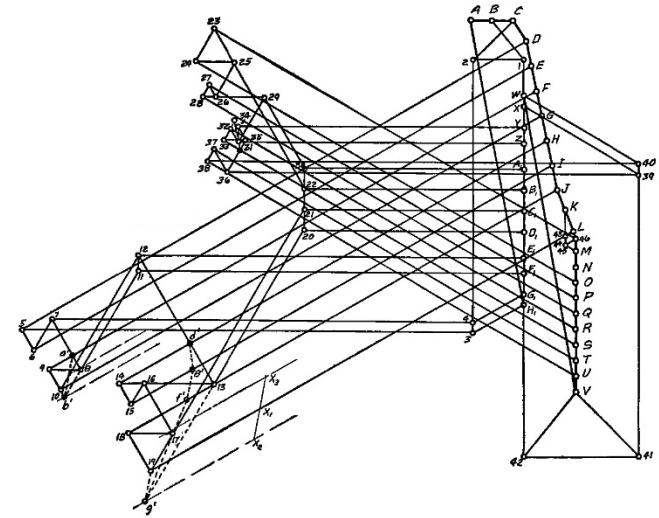








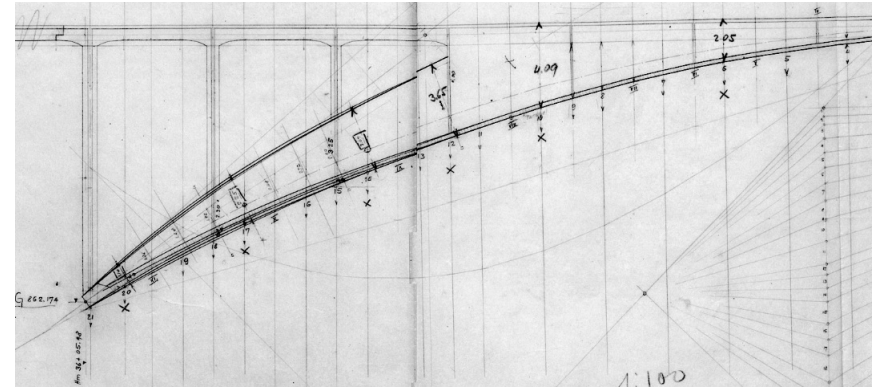
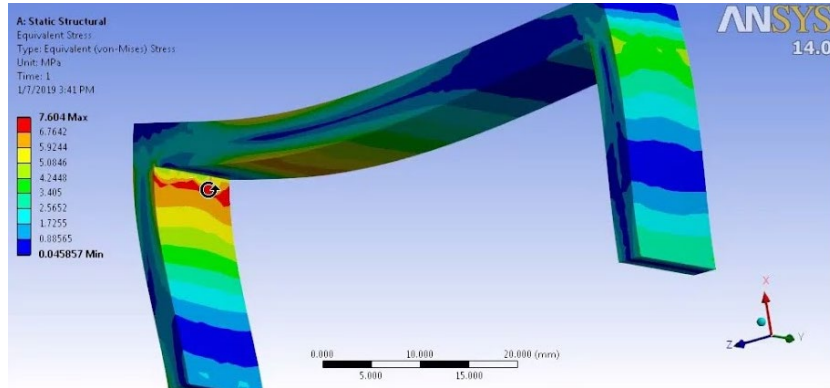
Form Diagram

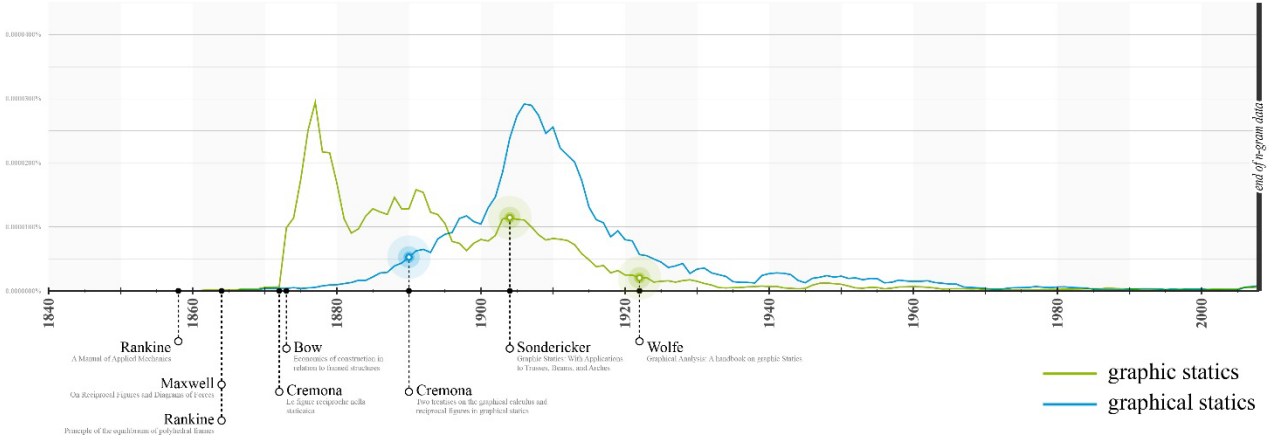
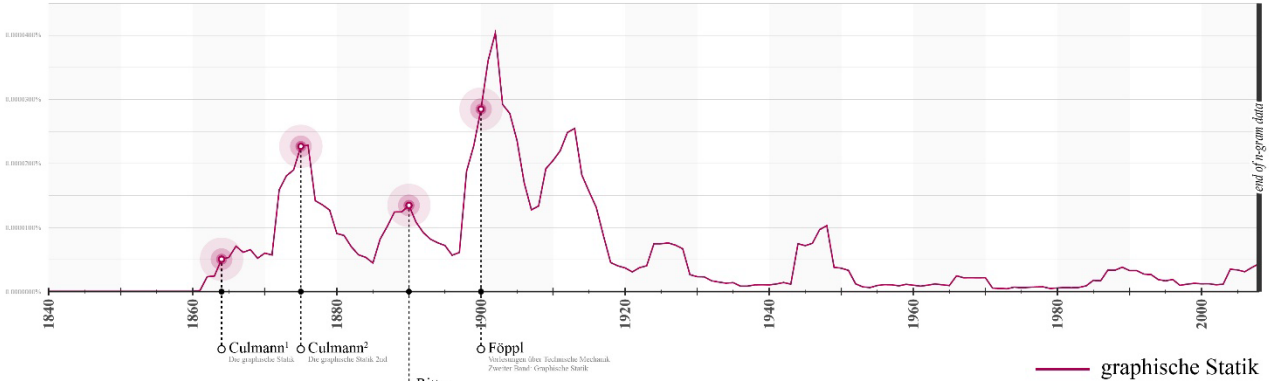


Force Diagram

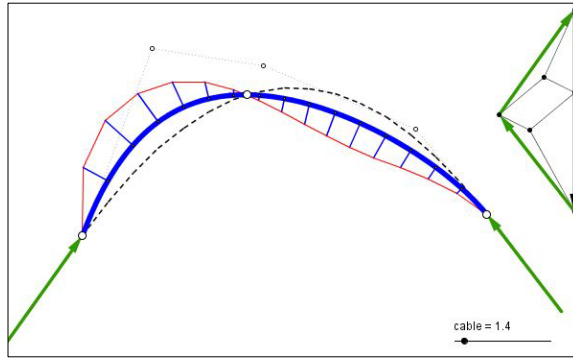
$$P = 24,000 \text{ lb}$$

$$= \underline{\underline{24^k}}$$

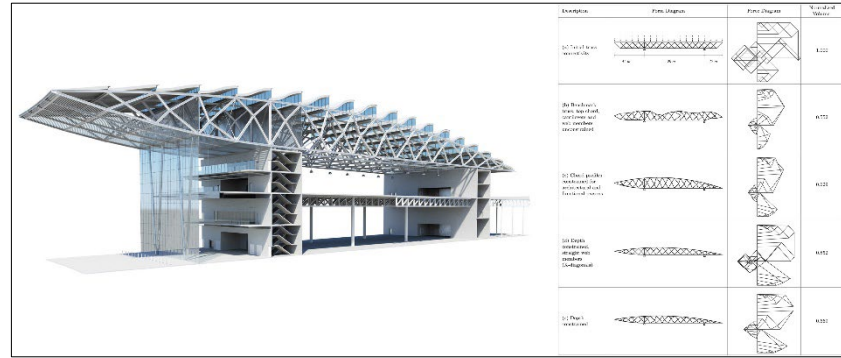






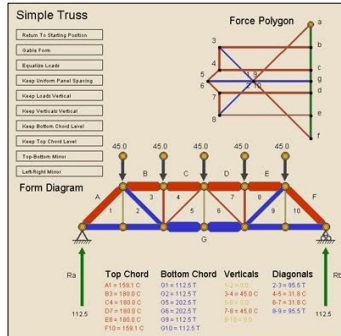
eEQUILIBRIUM

Van Mele et al. (2009-)



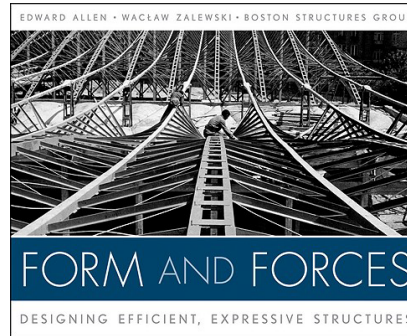
Structural Optimization using Graphic Statics

Beghini, et al. & SOM (2013)



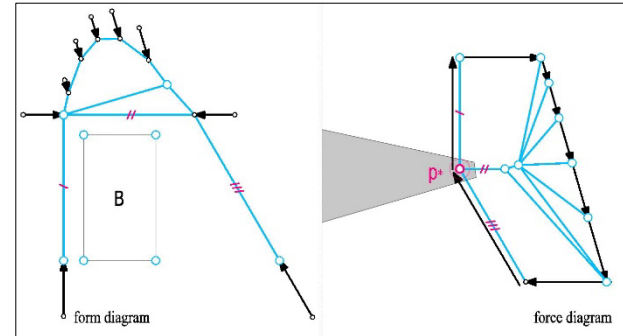
Active Statics

Greenwold & Allen (2003)



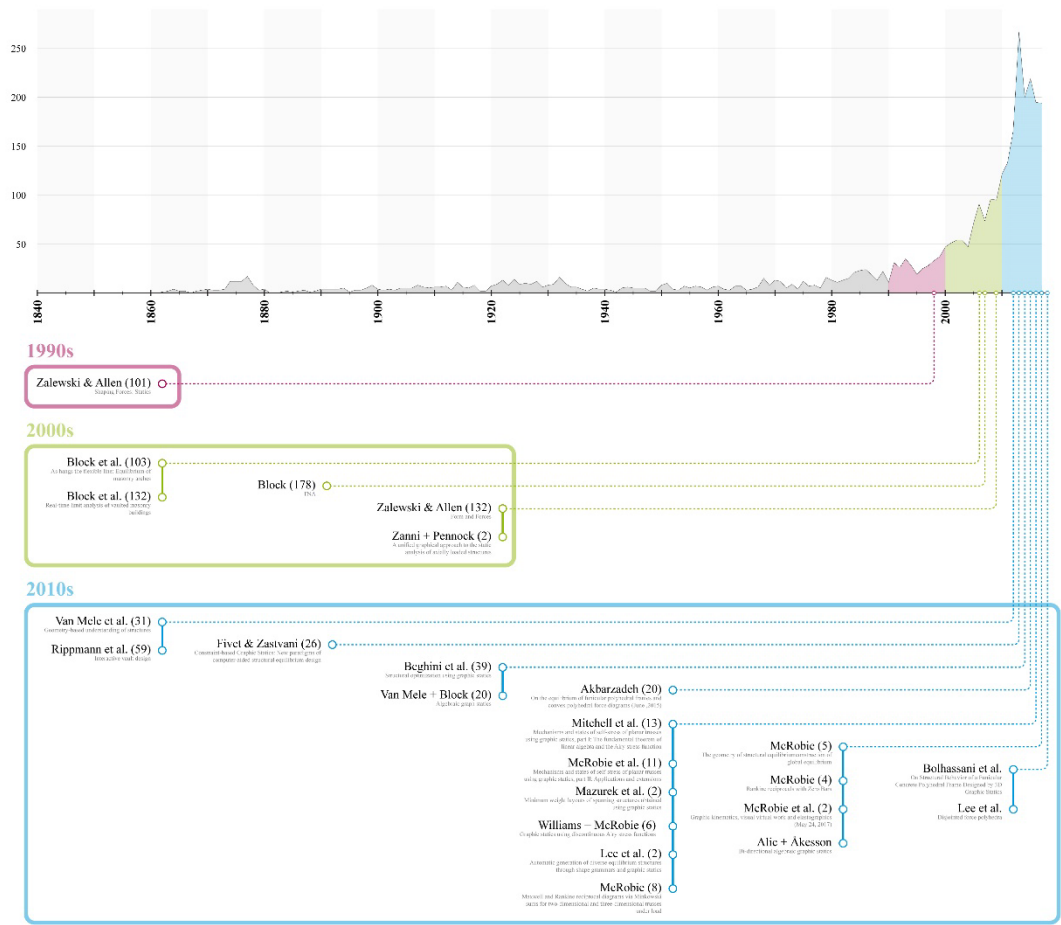
Form and Forces

Allen & Zalewski (2009)

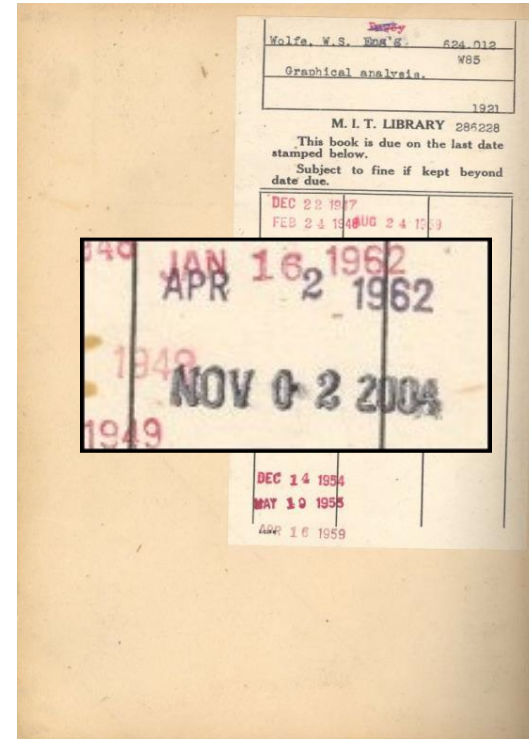
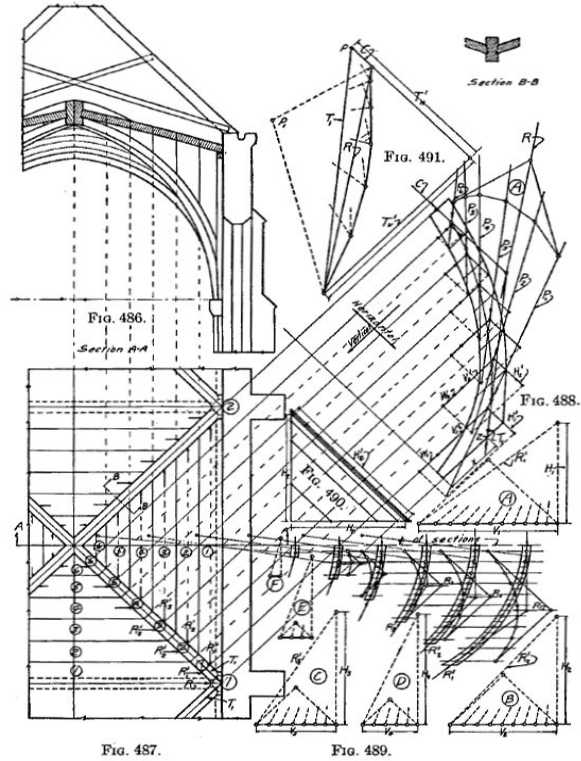


Constraint-based Graphic Statics

Fivet & Zastavni (2013)



Timeline of graphic statics publications (early 2018) | from PhD dissertation by Lee (2018)









PROFESSOR DR
CARL CULMANN
VON 1855 BIS 1881
VORSTAND DER
INGENIEURSCHULE
DES EIDGENÖSSISCHEN
POLYTECHNIKUMS



Carl Culmann

- Founder of graphic statics: “Die graphische Statik” (1864-1866)
- First Professor of Civil Engineering at the ETH Zurich (1855-1883)
- Director of the Polytechnic (1872-1875)

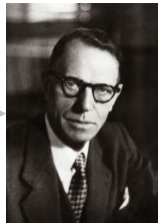


Karl Wilhelm Ritter (ETHZ diploma, 1868)

- Professor of graphic statics and bridge construction (appointed in 1882)
- Director of the Polytechnic (1887-1891)
- Extended applications of Culmann’s graphic statics:
“Anwendungen der graphischen Statik nach Professor Dr. Culmann” (1888-1906)



Robert Maillart
(ETH diploma, 1894)



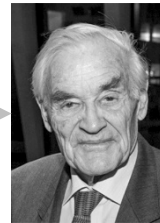
Othmar Ammann
(ETH diploma, 1902)



Pierre Lardy
(ETH diploma, 1928)



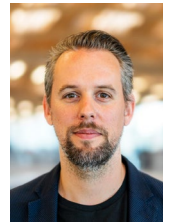
Heinz Isler
(ETH diploma, 1950)



Christian Menn
(ETH diploma, 1950)



Joseph Schwartz
(ETH professor, 2008-)



Philippe Block
(ETH professor, 2009-)

